MILITARY SPECIFICATION

FIXANT, HOLD DOWN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a film-forming liquid hold fixant. The hold down fixant solution shall consist of compounds as required, to conform to the requirements specified herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallets, Material Handling Wood Stringer Construction 2-Way and 4-Way (Partial).
RR-S-366	- Sieve, Test.
TT-E-489	- Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces).
VV-G-1690	- Gasoline, Automotive, Leaded or Unleaded.
PPP-P-704	- Palls-Metal: (Shipping, Steel, 1 through 12 Gallons).

MILITARY

MIL-F-24385	- Fire Extinguishing Agent, Aqueous Film-Forming Foam, (AFFF) Liquid Concentrate, for Fresh and Sea Water.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document shall be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-147 - Palletized Unit Loads.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government document. The following other Government. document forms a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION

Code of Federal Regulations, Title 49, Part 178.19

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 445 - Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity), Test. for.
D 522 - Elongation of Attached Organic Coatings with Conical Mandrel Apparatus, Test Method for. (DoD adopted)
D 609 - Preparation of Steel Panels for Testing Paint, Varnish, Lacquer, and Related Products.
D 1141 - Substitute Ocean Water, Specification for.
D 1331 - Surface and Interfacial Tension of Solutions of Surface-Active Agents, Tests for.
E 7 0 - pH of Aqueous Solutions with the Class Electrode, Test for. (DoD adopted)
G 2 1 - Determining Resistance of Synthetic Polymeric Materials to Fungi, Practice for. (DoD adopted)
G 2 2 - Determining Resistance of Plastics to Bacteria, Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103,)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among tchnical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflicct between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish sample units for first article inspection (see 4.3 and 6.3).

3.2 Materials. The fixant (see 6.4.1) shall be a water-based mixture conforming to the requirements specified herein.

3.2.1 Toxicity. The fixant shall have no adverse effect on the health of personnel when used for its intended purpose. The fixant shall be certified as safe (see 4.5).

3.2.2 Identification of samples. The first article sample submitted by the contractor shall be identified for the purpose of comparing with material subsequently submitted for acceptance. Within experimental limits, first article and quality conformance inspection samples shall be identical. Any lot or lots found not to be identical with the first article sample shall be rejected.

3.2.3 Composition. The composition (see 6.4.3) shall be appropriate formulations of acrylic polymers and dilute ammonium hydroxide solution to meet the performance and physical characteristics as specified herein. The use of infra-red spectroscopy (see 4.7.1) shall be used as the criteria for compositional control. 'The spectra of first article and quality conformance inspection samples shall be identical with respect to the number and location of infra-red absorption peaks. The peak heights shall not vary more than plus or minus 10 percent of that value determined for the first article inspection sample.

3.3 Hold down fixant characteristics. The hold down fixant shall conform to the chemical and physical requirements shown in table I.

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TABLE I. Chemical and physical requirements for fixant.

Requirement	Value	Application publication	Test paragraph
Viscosity, pascal seconds, maximum (max) at 20 ± 0.1°C	0.01	ASTM D 445	4.7.2
pH value 1/	7.0 to 9.0	ASTM E 70	4.7.3
Percentage of solids, minimum (min)	6		4.7.4
Surface tension, dynes per centimeter, max	45	ASTM D 1331	4.7.5
Elongation	no cracking	ASTM D 522 D 609	4.7.6
AFFF compatibility, burnback resistance, seconds, min	320	MIL-F-24385	4.7.12

1/ pH shall not vary by more than plus or minus 0.5 pH units from the value determined in the first article inspection when tested, as specified (see 4.7.3).

3.3.1 Storage. The hold down fixant shall conform to the following requirements after 10 days at 60 degrees Celsius (“C) (see 4.7.7).

(a) Viscosity (see table I).
(b) pH value (see table I).
(C) Volubility: as specified (see 3.3.3).
(d) Transparency: as specified (see 3.3.5).
(e) Surface tension (see table I).
(f) Elongation (see table I).
(g) Homogeneity: as specified (see 3.3.8).

3.3.2 Fixant performance compatibility. The hold down fixant shall be compatible with:

(a) Aqueous film-forming foam (AFFF): as specified (see 4.7.8.1).
(b) Sea water: as specified (see 4.7.8.2).

30303 Volubility. The film formed by the drying of the hold down fixant shall be insoluble in water up to 45 + 2°C with a pH value between 4.0 and 8.0, when tested, as specified (see 4.7.9).

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3.3.4 Strippability. The dry hold down fixant film shall be strippable (see 6.4.2) with a 5 percent aqueous solution of trisodium phosphate, when tested as specified (see 4.7.10).

3.3.5 Transparency. The dry film of the hold down fixant shall be colorless and transparent when tested as specified (see 4.7.11).

3.3.6 AFFF compatibility. The fixant solution shall be compatible with the fire-fighting agent AFFF. There shall be no degrading of AFFF performance, as specified in MIL-F-24385, when fixant is applied over an AFFF foam blanket and tested as specified (see 4.7.12).

3.3.7 Bacteria and fungal resistance. The hold down fixant shall resist the growth of bacteria or fungus, when tested as specified (see 4.7.13).

3.3.8 Homogeneity. Products under this specification shall be produced utilizing the proper blending of materials, to provide a mixture of uniform structure and composition throughout, as specified (see 4.7.14).

3.3.9 Unit content. The unit content shall be 5 gallons.

3.4 Marking.

3.4.1 Identification marking shall be In accordance with MIL-STD-130. In addition, the marking on the containers (see 5.3) shall be in black characters against a white background (see 5.1.1.3).

3.4.2 Two identical markings shall be applied to containers so that the markings are located diametrically opposite to each other. The markings shall be applied on the containers, so normal handllng will not impair the legibility of the marking. Paper labels shall not be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract , the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such Inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Examination of filled containers (see 4.6).

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4.2.1 Inspection conditions. Unless otherwise specified, all first article and quality conformance inspections shall be performed in accordance with the test conditions specified (see 4.5, 4.6, and 4.7).

4.3 First article inspection. First article inspections shall consist of those tests listed in table II.

TABLE 11. First article inspections.

Examination	Reference paragraph	
	Requirement	Test
Toxicity	3.2.1	4.5
Infra-red spectroscopy	3.2.3	4.7.1
Viscosity	3.3	4.7.2
pH value	3.3	4.7.3
Percentage of solids	3.3	4.7.4
Surface tension	3*3	4.7.5
Elongation	3*3	4.7.6
Storage	3.3.1	4.7.7
Fixant performance compatibility	3.3.2	4.7.8
Volubility	3.3.3	4.7.9
Strippability	3.3.4	4.7.10
Transparency	3.3.5	4.7.11
AFFF compatibility	3.3.6	4.7.12
Resistance to bacteria and fungal	3*3.7	4.7.13
Homogeneity	3.3.8	4.7.14
Examination of filled containers	3.4	4.6

4.3.1 Samples for first article inspection. Three 5 gallon containers (15 gallons) from each lot, as specified (see 4.4.1), are required for the first article inspection.

4.4 Quality conformance inspection. Quality conformance inspections shall consist of those tests listed in table 111.

TABLE III. Quality conformance inspections.

Examination	Reference paragraph	
	Requirement	Test
Infra-red spectroscopy	3.2.3	4.7.1
viscosity	3.3	4.7.2
pH value	303	4.7.3
Percentage of solids	3.3	4.7.4
Strippability	3*3.4	4.7.10
Transparency	3.3.5	4.7.11
Homogeneity	3.3.8	4.7.14
Examination of filled containers	3.4	4.6

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4.4.1 Inspection lot. A lot shall consist of all hold down fixant manufactured as a mass batch or job lot batch production, to assure required quality levels under large and small lot purchasing.

4.4.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected from each lot as specified (see 4.4.1), in accordance with MIL-STD-105 at inspection level I. The acceptable quality level (AQL) of 2.5 percent defective shall be used to verify compliance with all requirements regarding fill, closure, marking, and the requirements specified (see 4.6, 5.1.1.1 and 5.1.1.2).

4.4.3 Sampling for quality conformance inspection. Three filled 5 gallon containers shall be selected at random from each lot as specified (see 4.4.1) and used as one composite sample for the tests specified (see 4.7). The results of the tests required (see 4.4) shall be submitted to NAVSEA or the designated laboratory.

4.4.4 Quality conformance inspection report. The contractor shall prepare test reports in accordance with the data ordering document Included in the contract (see 6.2.2). If the sample tested is found not to be in conformance with any of the quallty conformance tests, the lot represented by the sample shall be rejected.

4.5 Toxicological data and formulations. The contractor shall furnish the toxicological data and formulations in accordance with the data ordering document included in the contract (see 6.2.2) to evaluate the safety of the material for the proposed use. Data and formulations shall be referred by the contracting activity to the Bureau of Medicine and Surgery (BUMED), to be certified as safe for use when requested by BUMED or its agent.

4.6 Examination of filled containers. Each sample filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects or less than required fill, shall not be offered for delivery. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, there shall be cause for rejection of the lot represented by the sample.

4.7 Test procedures.

4.7.1 Infra-red spectroscopy.

4.7.1.1 Procedure. Cut a 65 millimeters (mm) square of aluminum foil of .08 mm thickness (minimum) having one side mirror bright 1/. Wash the square with absolute alcohol, and air dry. The mirror bright side should give a spectrum that differs less than 2 percent transmittance from the standard front surface test

1/ Such a foil is available from Alu-Foil Co., 1143 Conklin St., Farmingdale, L.I., NY 11735.

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mirror, used in the specular reflectance accessory. Apply approximately 1 milliliter (mL) of fixant with an eye dropper in a zig-zag pattern, and produce a uniform layer by means of a 50 mm doctor blade with 0.02 mm gap. The sample thus prepared is allowed to dry overnight and then conditioned at a relative humidity of 50 + 5 percent at 25 ± 5°C for 24 hours. CAUTION: To the extent possible the film thickness should be the same for all samples to insure consistent. acceptable results.

4.7.1.2 Results. The spectrum of the coated foil is run on an infra-red spectrophotometer equipped with a specular reflectance accessory. The base line of the spectrum should show approximately 90 percent transmittance, and the most intense band should show approximately 10 percent. It may be necessary to move the coated aluminum foil slightly, to approach these conditions as nearly as possible.

4.7.2 Viscosity. The viscosity shall be determined at temperatures of . 20 ± O.l°C in accordance with ASTM D 445. The hold down fixant shall have a maximum dynamic viscosity of 10 centipoise (0.01 pascal seconds) at 20 ± O.l°C.

4.7.3 pH value. The pH value shall be determined potentiometrically, using a pH meter with a glass electrode. The hold down fixant shall have a pH value between 7.0 and 9.0, as determined according to ASTM E 70. Solution pH shall not vary more than plus or minus 0.5 pH units in any subsequent "quality assurance test sample'", In accordance with the data ordering document included in the contract.

4.7.4 percentage of solids. One determination of percentage solids will be made for each 5 gallon container. Determine weight of petri dish (120 x 20 mm). Pour approximately 50 grams (g) of fixant into petri dish and weigh again. Dry in convection oven at 50"C for 24 hours, cool in desiccator and weigh again. Repeat as required to obtain constant weight (± 0.05 g between weighings). Return petri dish to desiccator for 24 hours and weigh again. Percentage of solids shall be a minimum of 6 percent as calculated using the following formula:

$$\text{\% solids by weight} = \frac{(D+C-2A)100}{2(B-A)}$$

where A = weight of petri dish.

B = weight of petri dish with approximately 50 g fixant.

C = weight after drying at 50°C for 24 hours in convection oven and cooling in desiccator.

D = weight after replacing petri dish in desiccator for 24 hours.

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4. 7.5 Surface tension. The surface tension of the hold down fixant shall be determined by method A of ASTM D 1331.

4.7.6 Elongation.

4.7.6.1 Test specimen. A test shall be conducted on 3 steel panels in accordance with ASTM D 522. Spray hold down fixant onto one side of horizontal cold rolled steel panels measuring 76 x 152 x 0.9 mm. Steel panels shall be prepared as in ASTM D 609, method C or D prior to fixant application. Dry coated panels in convect-ion oven at 50°C for 24 hours, then condition panels at. relative humidity (RH) of 50 + 5 percent at 25 + 5"C for 24 hours. The film thickness after conditioning shall be equivalent to that expected in fixant use plus or minus 1.0 mm, but shall in no instance be less than 0.08 mm. Multiple sprayings may be necessary to obtain the minimum film thickness. A magnetic film thickness gage with at least an accuracy of 0.03 mm shall be used for the film thickness determinations. Film thickness shall vary no more than 0.03 mm, when measurements are taken in the centers of the four test panel quadrant-s.

4.7.6.2 Procedure. Test specimens shall be tested for film thickness and elongation immediately upon removal from conditioning apparatus. Film elongation shall be determined for three test panels using the apparatus and procedures of ASTM D 522. Film thickness measurements required for elongation calculations shall be performed as specified (see 4.7.6.1).

4.7.7 Storage.

4.7.7.1 Sample preparation. Samples of the hold down fixant shall be drawn from sample lots as specified (see 4.3.1). Two liters (L) from each 5 gallon container shall be placed in lightly stoppered glass cylinders and labeled for identification. All samples shall then be stored at 60°C for a period of 10 days. The samples shall then be subjected to the following tests:

(a) Viscosity (see table I).
(b) pH value (see table I).
(C) Volubility: as specified (see 3.3.3).
(d) Transparency: as specified (see 3.3.5).
(e) Surface tension (see table I).
(f) Elongation (see table I).
(g) Homogeneity: as specified (see 3.3.8).

4.7.8 Fixant performance compatibility.

4.7.8.1 Aqueous film-forming foam (AFFF).

4.7.8. 1.1 Procedure. Prepare synthetic sea water as specified in ASTM D 1141. Take 1 L of the synthetic sea water and use to formulate a 6 percent AFFF solution as specified in MIL-F-24385. Weigh out 5 g of 6 percent AFFF sea water solution into each of three 50 mL beakers. Add 100 g hold down fixant into each of the three beakers. All solutions shall remain free of coagulation or precipitation. Samples showing degradation of the hold down fixant will be rejected.

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4.7.8.2 Synthetic sea water.

4.7.8.2.1 Procedure. Prepare synthetic sea water as specified in ASTM D 1141. Weigh out 5 g of synthetic sea water into each of three 50 mL beakers. Add 100 g hold down fixant into each of the three beakers. All solutions shall remain free of coagulation or precipitation. Samples showing degradation of the hold down fixant will be rejected.

4.7.9 Volubility.

4.7.9.1 Test specimen. Pour a quantity of hold down fixant in a petri dish (120 x 20 mm) which will form a 0.15 - 0.25 mm film. Dry in convection oven at 50°C for 24 hours, then place in a desiccator for 24 hours at a RH of 50 ± 5 percent at 25 + 5°C.

4.7.9.2 Distilled water.

4.7.9.2.1 Procedure. Use two films prepared as specified (see 4.7.9.1). Fill a beaker with 250 mL of distilled water. Heat water to 45 + 2°C. Pour 00 mL from the beaker on each film in petri dish. Wait 5 minutes, then drain ff water. Check film for dissolution. Film may discolor or distort due to the moisture, but it should not dissolve.

4.7.9.3 Synthetic sea water.

4.7.9.3.1 Procedure. Use two films as specified (see 4.7.9.1). Fill a beaker with 250 mL of synthetic sea water. Heat to 45 + 2°C. Pour" 100 mL from beaker on each film in petri dish. Wait 5 minutes, then drain off water. Check film for dissolution. Film may discolor or distort due to the moisture but It should not dissolve.

4.7.10 Strippability.

4.7. 10.1 Test specimen. Use two films prepared as specified in 4.7.9.1.

4.7. 10.2 Procedure. Prepare 5 percent aqueous solution of trisodium phosphate and adjust temperature to 45 + 2°C. Use two films prepared previously. Pour 100 mL of stripping solution onto each film. Wait 5 minutes, then drain off the solution. Film should have dissolved sufficiently for easy removal from petri dish with flushing.

4.7.11 Transparency.

4.7. 11,1 Test specimen. Use two films prepared as specified in 4.7.9.1.

4.7.11.2 Procedure. Inspect each of the prepared films for clarity and discoloration. Samples should be clear with no discoloration, A very slight blue r yellow tinge is acceptable.

4.7.12 AFFF compatibility. The fixant's compatibility with AFFF liquid concentrate , 6 percent, shall be determined by measuring the burnback time in the presence of the hold down fixant.

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4.7.12.1 Test equipment. The fire test shall be conducted in a level, circular pan 1.8 m in diameter, fabricated from 6 mm thick steel with a 100 mm high side. A shallow water layer shall be used to protect the pan bottom and to ensure complete coverage of the area with fuel.

4.7.12.2 Test materials. The fuel shall be gasoline conforming to VV-G-1690. The AFFF shall conform to MIL-F-24385. The seive shall be a 200 mm diameter, 40 mesh sieve conforming to type I, style A of RR-S-366.

4.7.12.3 Test procedure. Ten gallons of fuel shall be dumped within a 30-second period. The fuel shall be ignited within 30 seconds of fueling and allowed to burn freely for 10 seconds. After the preburn period, the fire shall be extinguished expeditiously and the fire extinguishing time shall be recorded at the exact cessation of all flames, but AFFF application shall continue for a total of 90 seconds. The application foam rate shall be as defined for the 2.4 meter square (m2) fire test in accordance with MIL-F-24385. Within 60 seconds of the completion of AFFF application, 1 L of hold down fixant shall be evenly sprayed over the foam blanket. A burning pan (0.3 m in diameter with 50 mm side) shall be placed in the center of the 2.6 (m^2) pan and a time started. When it appears that the fire has spread outside Lhe pan so that burning will continue after pan removal, the pan shall be removed. The burnback time is that time at which it is estimated that 0.6 m2 (25 percent) of the total area is involved in flames.

NOTE : Intermittent "flash-overs" may occur. They are characterized by creeping faint blue or invisible flames over the foam surface which usually self-extinguish. They are not considered a part of the burnback area unless sustained burning occurs. All isolated, sustained burning areas shall be included in arriving at the 0.6 m2 total area.

4.7.13 Bacteria and fungal resistance. The hold down fixant shall be tested for resistance to bacterial or fungal growth as specified in ASTM G 21 and ASTM G 22.

4.7.14 Homogeneity. The proper blending of materials to provide a mixture of uniform structure and composition throughout, shall be determined by visual examination of the samples contained in the glass beakers.

4.7.14.1 Procedure. Fill a clear 100 mL glass graduate with fixant. Look through the longitudinal axis of the cylinder and fixant at a light source.

4.7. 14.2 Results. The mixture shall appear uniform in consistency and be free of lumps, sediment, dirt, foreign material, and gelatinous or colloidal precipitate.

4.7.15 Torque test (pour cap of 5 gallon plastic container). The pour cap of the 5 gallon plastic container shall be subjected to a torque test to determine conformance with 5. 1.1.1.1 (f). This test shall be performed a minimum of 48 hours after initial closure and torquing of the container cap. The time clement will permit the phenomena of plastic "cold flow" to take effect.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the ?reservat-ion-packaging, packing and marking for shipment and storage

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shall be in accordance with the requiretnents of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct government acquisitions.)

5.1 Preservation. Preservation shall be level A as specified hereinafter.

5.1.1 The hold down fixant shall be furnished in a 5 gallon plastic or metal pall as specified (see 6.2.1).

5.1.1.1 Plastic pails. The pail shall be molded polyethylene as specified herein.

Capacity	5 gallon (minimum)
Height, body (overall)	14-1/4 inches (+ l/4-inch)
Diameter, body (overall)	11-1/2 inches (+ 3/8 - 1/8 inch)
Pour opening (inside dimensions)	1-1/2 inches (minimum)

5. 1.1.1.1 The pail shall meet the requirements of Department of Transportation Specification Number 34 as specified in the Code of Federal Regulations, Title 49, Part 178.19, and as follows:

(a) Shall be stackable and self-supporting.
(b) Shall be provided with a threaded type plastic cap fitted with a gasket for the pour opening.
(c) May be provided with a vent opening having an easily punctured membrane. When furnished, vent opening shall be provided with a threaded type plastic cap.
(d) Shall be provided with an integrally molded or recessible plastic or metal handle.
(e) Shall have a white color conforming to S. 1.1.3.
(f) The torque required to remove the pour opening cap shall not exceed 50 inch-pound (see 4.7.15).

5.1.1.2 Metal pails. Pails shall conform to class 2 or 5, type I of PPF-P-704. Pails shall be provided with press-in type plastic nozzle closure with plastic push-pull spout. The pour opening shall have a minimum diameter of 1-1/2 inches. Wire handles shall be galvanized or protectively coated, to resist corrosion,

5.1.1.3 Exterior color and coatings. Color shall be white and shall be an approximate match to color number 27875 of FED-STD-595. Exterior coating for metal pails shall conform to TT-E-489. In case of question or dispute, a color chip shall be submitted to the contracting activity for resolution.

5.2 packing. Levels A, B and C material packaged as specified shall be palletized.

5.2.1 Palletization. Thirty-six 5 gallon plastic or metal pails shall be palletized in accordance with load type XVIII of MIL-STD-147. Pallets conforming

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to type V, class 1, wood group optional, size 2 of NN-P-71, are acceptable. Containers shall be properly and firmly nested and arranged to insure a snug, non-shifting load. pallet dimensions may be adjusted to assure a snug, non-shifting load , but shall not exceed 43 x 52 inches.

5.3 Marking. In addition to the marking specified (see 3.4) and any special marking required (see 6 .2.1), containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The fixant is intended for use in mechanical spray generating equipment such as spray containers for containing potential release of carbon fibers during carbon composite fire clean-up operations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) First article sample (see 3.1, 4.3.1).
(c) Type of container required (see 5.1.1).
(d) Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered In accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4.4	Reports, test	DI-T-2072	10.1.b
4.5	Certificate of compliance	DI-E-2121	--

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such In DoD 5000. 19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of this specification required to be performed Lo meet a data requirement may be waived by t-he contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract., regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Hold down fixant or fixant. A material to be applied to burned composite debris, to prevent or reduce the loss of carbon fiber to the atmosphere.

6.4.2 Strippability. The ability of a fixant material to be removed from the composite material by washing.

6.4.3 Carbon composite. Structural material consisting of carbon fiber or cloth impregnated with a binder such as epoxy resin.

Custodian:
Navy - SH
Air Force - 99
Review activities:
Navy - AS, MC, YD
DLA - CS

Preparing activity:
Navy - SH
(project 4210-0345)

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seawater temperature (°F)

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OMB Approval
No. 22-R255

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MIL-F-24610

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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